OIML Certificate

OIML Member State
The Netherlands

Number R60/2021-A-NL1-26.18 revision 0
Project number 4164804

Issuing authority	NMi Certin B.V. Person responsible: M.Ph.D. Schmidt
Applicant and Manufacturer	Qlar Europe GmbH Pallaswiesenstrasse 100 64193 Darmstadt Germany
Identification of the certified type	A **compression load cell**, with strain gauges Registered trade name : Qlar Europe GmbH Type : RTB SENSiQ RTB SENSiQ Ring-Torsion Load Cell RTB
Characteristics	See next page

This OIML Certificate is issued under scheme A.

This Certificate attests the conformity of the above identified Type (represented by the sample(s) identified in the OIML Test Report) with the requirements of the following Recommendation of the International Organization of Legal Metrology (OIML):

OIML R 60-1:2021 for accuracy class C

This Certificate relates only to the metrological and technical characteristics of the type of measuring instrument covered by the relevant OIML International Recommendation above-identified.
This Certificate does not bestow any form of legal international approval.

This certificate and supporting reports comply with the requirements of OIML-CS-PD-07 clause 6.2.

Important note: Apart from the mention of the Certificate's reference number and the name of the OIML Member State in which the Certificate was issued, partial quotation of the Certificate and of the associated OIML Test Report(s) is not permitted, although either may be reproduced in full.

Issuing Authority

NMi Certin B.V., OIML Issuing Authority NL1
3 September 2026

Certification Board

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This document is issued under the provision that no liability is accepted and that the applicant shall indemnify third-party liability.

The notification of NMi Certin B.V. as Issuing Authority can be verified at www.oiml.org

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The conformity was established by the results of tests and examinations provided in the associated reports:

- No. NMi-10200947-01 dated 24 December 2010 that includes 59 pages;
- No. NMi-11200444-03 dated 24 August 2011 that includes 65 pages;
- No. NMi-11200444-04 dated 24 August 2011 that includes 59 pages;
- No. NMi-11200444-05 dated 24 August 2011 that includes 59 pages.

Characteristics of the load cell:

Characterization of load cell capabilities	Analog-passive load cell		
Maximum capacity (E_{max})	60 kg, 130 kg, 250 kg, 280 kg	500 kg up to and including 10 t	13 t up to and including 60 t
Minimum dead load	0 kg		
Accuracy Class	C		
Rated Output	1,0 mV/V ± 0,1 mV/V for 60, 130 and 280 kg 1,75 mV/V ± 0,1 mV/V for 250 kg 2,0 mV/V ± 0,1 mV/V for 500 kg up to and incl. 60 t		
Maximum number of load cell intervals (n) (1)	5000		
Ratio of minimum LC Verification interval (1) $Y = E_{max} / v_{min}$	16000	17500	25000
Ratio of minimum dead load output return (1) $Z = E_{max} / (2 * DR)$	5000		
Input impedance	1260 Ω ± 100 Ω for 60, 130 and 280 kg 1100 Ω ± 100 Ω for 250 kg up to and incl. 10 t 1200 Ω ± 100 Ω for 13 t 1075 Ω ± 100 Ω for 28 t 1350 Ω ± 200 Ω for 60 t		
Temperature range	-10 °C / +40°C		
Fraction p_{LC}	0,7		
Humidity Class	CH		
Safe overload	150 % of E_{max}		
Output impedance	1020 Ω ± 0,5 Ω for 60, 130 and 280 kg 1025 Ω ± 25 Ω for 250 kg up to and incl. 10 t 1000 Ω ± 0,5 Ω for 13 t 930 Ω ± 0,5 Ω for 28 t 1175 Ω ± 0,5 Ω for 60 t		
Recommended excitation	5-12 V AC / DC		
Excitation maximum	18 V AC / DC		

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Transducer material	Stainless steel
Atmospheric protection	Hermetically welded

Remark:

1. The characteristics for n_{max}, Y and Z can be reduced separately.

Each load cell produced is provided with an accompanying document with information about its characteristics.

Revision History

Revision	Date	Change(s)
0	2026-09-03	Initial issue.